Proxmox VE Hypervisor Deployment

Overview

This document covers the deployment of a Proxmox VE hypervisor node into the existing lab infrastructure, including VLAN-aware networking configuration, trunk port migration, and integration with the Management VLAN.

A dedicated desktop machine was added to the lab as a bare-metal hypervisor running Proxmox VE. The node is assigned to the Management VLAN at `192.168.10.6` and connected to a trunk port on the TP-Link TL-SG108E switch, giving it simultaneous access to all six lab VLANs through a single NIC.

This enables virtual machines hosted on Proxmox to be placed on any lab VLAN by setting a VLAN tag at the VM network interface level, without requiring separate physical connections per VLAN.

Node Specifications

Property	Value
Hostname	proxmox-01
IP Address	192.168.10.6
VLAN	10 (Management)
Platform	Proxmox VE
NIC	Single NIC (nic0)
Switch Port	Port 2 (Trunk, all VLANs tagged)
Web UI	https://192.168.10.6:8006

Network Architecture

Why a Trunk Port

A single NIC carrying multiple VLANs requires an 802.1Q trunk port on the switch side and a VLAN-aware bridge on the host side. Without this, a hypervisor is limited to the single VLAN assigned to its switch port.

With a trunk port and a VLAN-aware Linux bridge:

- The physical NIC carries all VLAN traffic simultaneously, tagged
- The bridge decodes the 802.1Q tags and routes traffic to the correct VM
- Each VM is assigned to a VLAN by setting a VLAN Tag field at the virtual NIC level

- pfSense handles DHCP and routing for each VLAN as it already does for all other lab hosts

Switch Port Assignment

Port	Type	Connected To
Port 1	Trunk, all VLANs tagged	pfSense ue0 interface
Port 2	Trunk, all VLANs tagged	Proxmox VE (nic0)
Port 3	Access, VLAN 10	Secondary unmanaged switch
Ports 4–8	Access, individual VLANs	Reserved per VLAN

Port 3 feeds a secondary unmanaged switch that extends VLAN 10 connectivity to additional management-tier devices including Windows Server 2022 and the primary laptop.

Network Configuration

/etc/network/interfaces

```
auto lo
iface lo inet loopback

# Physical NIC - no IP, pure trunk carrier
auto nic0
iface nic0 inet manual

# VLAN-aware bridge - handles all VLANs through single physical NIC
auto vmbr0
iface vmbr0 inet manual
        bridge-ports nic0
        bridge-stp off
        bridge-fd 0
        bridge-vlan-aware yes
        bridge-vids 2-4094

# VLAN 10 sub-interface - Proxmox management IP lives here
auto vmbr0.10
iface vmbr0.10 inet static
        address 192.168.10.6/24
        gateway 192.168.10.1
        dns-nameservers 192.168.10.1

source /etc/network/interfaces.d/*
```

Interface Roles

Interface	Role	IP
nic0	Physical NIC, trunk carrier	None
vmbr0	VLAN-aware Linux bridge	None
vmbr0.10	VLAN 10 management sub-interface	192.168.10.6/24

Key Configuration Details

`bridge-vlan-aware yes` enables 802.1Q processing on the bridge. Without this, tagged frames arriving from the trunk port are not decoded and the host loses connectivity when moved to a trunk port.

`bridge-vids 2-4094` instructs the bridge to accept any VLAN tag in that range, covering all six lab VLANs and any future additions.

The management IP is placed on `vmbr0.10` rather than directly on `vmbr0`. On a trunk port, VLAN 10 traffic arrives tagged, the sub-interface explicitly handles VLAN 10 tagged frames, ensuring the management IP remains reachable after the trunk migration.

Deployment Procedure

Pre-requisites

- Proxmox VE ISO flashed to USB (available at proxmox.com/downloads)
- Static IP allocation confirmed: 192.168.10.6
- SSH access available during configuration
- Machine connected to a VLAN 10 access port for initial installation

Installation

During the Proxmox installer, configure the network as follows:

```
Management Interface:  [select physical NIC]
Hostname:              proxmox-01.homelab.local
IP Address:            192.168.10.6
Netmask:               255.255.255.0
Gateway:               192.168.10.1
DNS:                   192.168.10.1
```

Connect the machine to a VLAN 10 access port for the initial installation. The trunk migration is performed after installation is complete.

Trunk Port Migration

The trunk migration must follow this sequence exactly. Reversing the order causes immediate loss of access before the fix is in place.

Step 1, Verify the current NIC name

```
ip link show
```

Output of ip link show confirming nic0 as the physical NIC

Output of ip link show, the physical NIC is identified as nic0 with alternate names enp0s25 and enx3417eb9da246

Note the physical NIC name. On this node it is `nic0`. On other hardware it may differ, use whatever appears here, not an assumed name.

Step 2, Back up the existing config

```
cp /etc/network/interfaces /etc/network/interfaces.backup
```

Step 3, Review the default post-install config

```
cat /etc/network/interfaces
```

The default config assigns the management IP directly to `vmbr0` with no VLAN awareness. This works on an access port but will fail on a trunk port because tagged frames cannot be decoded.

Step 4, Edit the interfaces file

```
nano /etc/network/interfaces
```

New /etc/network/interfaces config in nano showing VLAN-aware bridge configuration

The updated interfaces file, vmbr0 is now VLAN-aware and the management IP has moved to vmbr0.10

Replace the contents with the configuration shown in the Network Configuration section above.

Step 5, Apply the configuration

```
ifreload -a
```

Proxmox uses `ifupdown2` which allows network changes to be applied without a reboot.

Step 6, Verify before moving the cable

```
# Management IP must be present on vmbr0.10
ip addr show vmbr0.10

# Must return 1 to confirm VLAN filtering is active
```

```
cat /sys/class/net/vmbr0/bridge/vlan_filtering

# Default route must point to pfSense via vmbr0.10
ip route
```

Verification command outputs after ifreload showing vmbr0.10 with correct IP and vlan_filtering returning 1

Verification output, vmbr0.10 holds 192.168.10.6/24, vlan_filtering returns 1, default route is correct

At this stage a ping to `192.168.10.1` returns `Destination Host Unreachable`. This is expected, the node is now sending tagged frames but is still on an access port that only accepts untagged frames. This confirms the config is working correctly and is ready for the trunk port.

Step 7, Move the cable

Move the Ethernet cable from Port 3 (access) to Port 2 (trunk) on the TP-Link switch. Wait ten seconds, then verify:

```
ping -c 4 192.168.10.1
```

Successful ping to pfSense after moving to trunk port showing 0% packet loss

4 packets transmitted, 4 received, 0% packet loss, Proxmox is live on the trunk port

Step 8, Confirm internet access

```
ping -c 4 8.8.8.8
```

Step 9, Confirm web UI

From any device on the Management VLAN:

```
https://192.168.10.6:8006
```

Proxmox VE web UI accessible at 192.168.10.6:8006

Proxmox web UI, the no-subscription notice is standard on the free community edition and does not affect functionality

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Creating VMs Across VLANs

With `vmbr0` VLAN-aware and on a trunk port, placing a VM on any lab VLAN requires only the VLAN Tag field in the VM network configuration.

In the Proxmox web UI when creating or editing a VM, set:

Field	Value
Bridge	vmbr0
VLAN Tag	Target VLAN ID

Examples:

VM	Bridge	VLAN Tag	Expected IP Range
Kali Linux (RedTeam)	vmbr0	30	192.168.30.50–100
Security tooling (BlueTeam)	vmbr0	20	192.168.20.50–100
CI/CD runner (DevOps)	vmbr0	40	192.168.40.50–100
Monitoring agent	vmbr0	60	192.168.60.50–100

pfSense automatically issues DHCP leases from the correct pool for each VLAN. No additional pfSense configuration is required beyond what is already in place.

Ansible Integration

Add Proxmox to the Ansible inventory from the controller node (192.168.10.2):

```
# Distribute the Ansible SSH key
ssh-copy-id -i ~/.ssh/ansible-homelab-key.pub root@192.168.10.6

# Add to inventory
echo "192.168.10.6   # Proxmox Hypervisor proxmox-01 - $(date)" >> /etc/ansible/hosts

# Verify connectivity
ansible 192.168.10.6 -m ping
```

Post-Deployment Verification Checklist

- [ ] Web UI accessible at https://192.168.10.6:8006
- [ ] `ip addr show vmbr0.10` shows 192.168.10.6/24
- [ ] `cat /sys/class/net/vmbr0/bridge/vlan_filtering` returns 1
- [ ] Ping to 192.168.10.1 succeeds
- [ ] Ping to 8.8.8.8 succeeds
- [ ] Ansible ping from controller (192.168.10.2) succeeds
- [ ] Test VM on VLAN 30 receives a 192.168.30.x address via DHCP

Troubleshooting

Lost connectivity immediately after ifreload

A syntax error in `/etc/network/interfaces` or the wrong NIC name was used. Restore the backup:

```
cp /etc/network/interfaces.backup /etc/network/interfaces
ifreload -a
```

Ping to pfSense fails after ifreload but before moving the cable

This is expected behaviour. The node is sending tagged frames to an access port. Proceed to move the cable to Port 2.

Ping to pfSense fails after moving the cable

Confirm Port 2 on the TP-Link switch is configured as a trunk port with all VLANs tagged. Access the switch web UI and verify the VLAN configuration for Port 2 matches Port 1.

Web UI unreachable but SSH works

```
systemctl restart pveproxy
```

Related Documentation

- 01-network-infrastructure.md, pfSense and switch VLAN configuration
- 04-automation-platform.md, Ansible controller setup
- 06-ansible-service-account.md, Service account configuration for new nodes
- 09-bootstrap-procedures.md, Standardised bootstrap procedures

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